higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY

(Second Paper)

NQF LEVEL 3

(10401023)

22 February 2019 (X-Paper)
09:00–12:00

Nonprogrammable calculators may be used.

This paper consists of 8 pages and 2 answer sheets.

NCV

TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

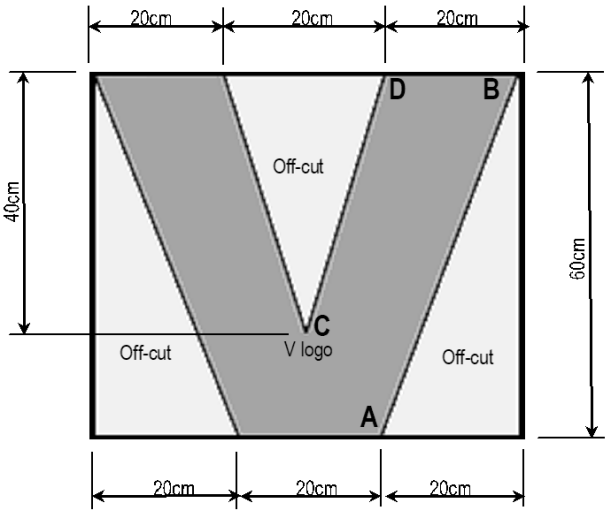
1. Answer ALL the questions.
2. Read All the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Clearly show ALL calculations.
6. Round off your answers correctly according to the given context. In all other cases, where the context is not specific, round off your answers correctly to two decimal places.
7. Indicate units of measurement where applicable.
8. Diagrams are NOT necessarily drawn to scale.
9. Answer QUESTION 3.2.3 on ANSWER SHEET A and QUESTION 4.8 on ANSWER SHEET B. Write EXAMINATION NUMBER in the spaces provided on the ANSWER SHEETS and hand them in with your ANSWER BOOK.
10. Write neatly and legibly.

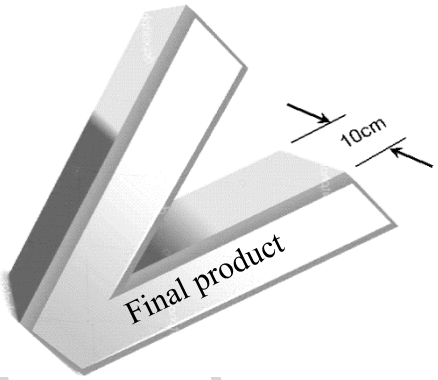
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QUESTION 1

Vusi Shabalala works for a designing company. He designed and produced a V-shaped logo for a client. The logo was produced from a wooden block. The thickness of the wooden block is 10 cm.

Below is a diagram showing the front view of the wooden block and a picture of the of the V-shaped logo.





Formulae:

Area of triangle= ½ × base × height.
Area of square = side × side
Pythagoras: 'The square on the hypotenuse = sum of the squares of the other two sides'

Study the above diagram and picture then answer the following questions:

1.1 Calculate the total area of the offcuts to produce the V-shaped logo. (9)

1.2 Calculate the area of the front view of the V-shaped logo. (6)

1.3 Use the theorem of Pythagoras to determine:

1.3.1 Length AB (2)

1.3.2 Length CD (2)

1.4 Hence, calculate the total surface area of the V-shaped logo. (9)

1.5 Determine the percentage of wood that was used to produce a V-shaped logo out of the wooden block. (3)

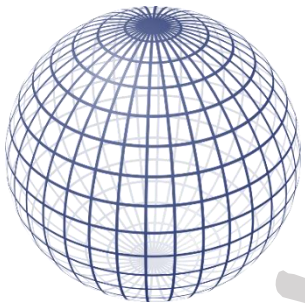
1.6 The V-shaped logo was spray painted. One tin of spray paint covers the surface area of 5 000 cm^2.



How many tins of spray paint are required to cover the surface area of 30 logos? (5)

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1.7 The client requested for the V-shaped logo to be placed inside a sphere with a diameter of 62 cm, like the one below:



Diameter = 62 cm

Calculate the volume of the sphere.

Formula: Volume of Sphere = $\frac{4}{3}\pi r^3$, where $\pi = 3{,}14$ (4)

[40]

QUESTION 2

2.1 Vusi Shabalala's salary advice for the month of February 2018 appears below. The missing values on the salary advice are labelled A, B, C and D.

SALARY ADVICE			
Employee name	**Employee no.**	**Job title**	**Salary level**
V Shabalala	CD2544	Design officer	SL 8C
Employee address: 4666 Moretlwa Street Nelmapius		Salary date Tax number Bank Account no.	28/02/2018 012244159 Blue Bank 112 456 458
EARNINGS		**DEDUCTIONS**	
Description	**Taxable**	**Description**	**Amount**
Basic salary	R22 611,00	PAYE	R6 827,54
Allowances (taxable)	R12 360,98	Pension fund	R1 695,87
		UIF	**B**
		Medical aid	R4 500,00
		Insurance	R450,00
		Union membership	R35,00
GROSS INCOME	**R A**	**TOTAL DEDUCTIONS**	**R C**
		NET INCOME	**R D**

2.1.1 PAYE is deducted from Mr Shabalala's salary. Why do employees pay towards PAYE? (1)

2.1.2 Mr Shabalala pays 1% of his basic salary towards UIF.

Calculate the value of B, Mr Shabalala's contribution towards UIF. (3)

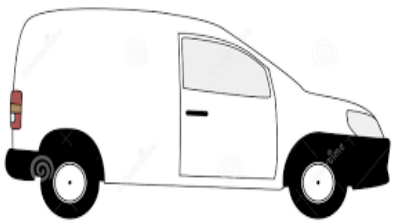
2.1.3 Calculate Mr Shabalala's net income (D).

Formula: Net income (D) = gross income (A) – total deductions (C) (5)

2.1.4 Mr Shabalala's basic salary increases to R24 193,77.

Calculate the percentage increase of his basic salary. (5)

2.2 Mr Shabalala wants to buy a new light delivery vehicle (LDV). The cost price of the LDV is R200 000. Vusi considers the option of buying the vehicle through a bank loan. He uses an online repayment calculator to consider possible monthly repayments. The table below was developed from information extracted from the online calculator.



It shows the:

- monthly repayment for every R1 000
- interest rate
- repayment period

Monthly repayment of a bank loan per R1 000			
Interest rate	9%	11%	13%
48 months repayment period	R24,89	R25,85	R26,83
72 months repayment period	R18,03	R19,03	R20,07

Study the above table and answer the following questions:

2.2.1 Determine the loan amount if Mr Shabalala must pay 10% towards a deposit. (3)

2.2.2 How many thousands are there in R180 000? (2)

2.2.3 Calculate the monthly repayment of R180 000 loan at 9% interest rate for a period of 48 months which includes a R57,00 monthly administration fee. (4)

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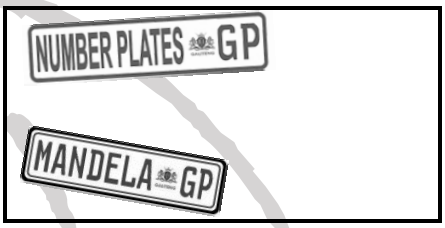
2.2.4 The bank charges an interest rate based on the customer's credit rating, gross income and other factors. Mr Shabalala qualifies for one of the following two options from the table:

Option 1: A loan of 180 000 for 48 months at 11% interest

Option 2: A loan of 180 000 for 72 months at 13% interest

Calculate the difference in the amount that will be paid at the end of the loan period between option 1 and option 2. Show ALL calculations. (12)

2.3 Mr Shabalala paid a VAT inclusive amount of R460,00 for number plates. VAT is charged at 15%.



Calculate the price of the number plates excluding VAT. (5)

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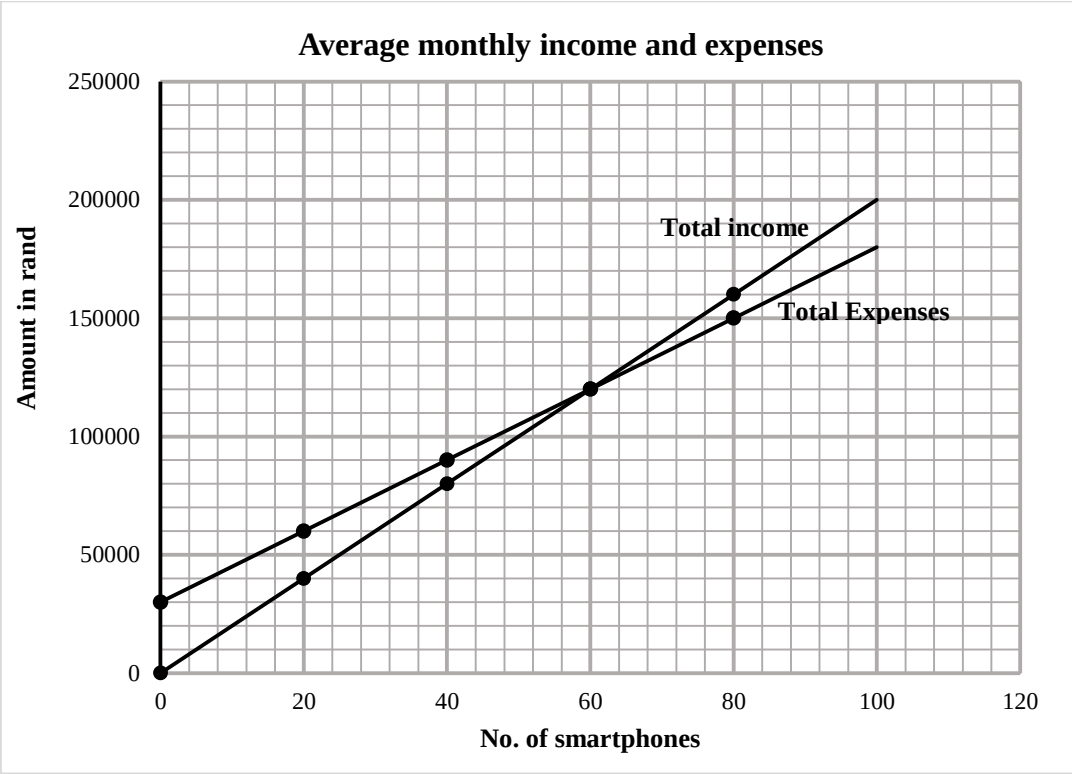
QUESTION 3

3.1 John orders low cost smartphones and sells them online in South Africa. The average total monthly expenses of John's business include a fixed monthly running cost and an average monthly ordering cost.

The fixed monthly running cost of the business include renting of offices and storage space, courier services for door-to-door deliveries, a website for marketing and processing of sales, telephone, water and lights, wages, etc. The average monthly cost of ordering these smartphones include custom fees, shipment, etc.

The straight-line graphs below show the average monthly income and expenses of John's business:

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Study the above graphs to answer the following questions:

3.1.1 How much does John pay towards the fixed monthly running cost of his business? (2)

3.1.2 Use calculations to show that the average cost to order one smartphone is R1 500. (3)

3.1.3 Derive a formula to calculate the average total monthly expenses of John's business. (3)

3.1.4 Use calculations to show that the average selling price of a smartphone is R2 000. (2)

3.1.5 Derive a formula to calculate the average total monthly income of John's business. (2)

3.1.6 What is the minimum number of smartphones that John must sell, per month, to make a profit? (2)

3.1.7 Use the graphs to calculate the profit that John will make if he sells 100 smartphones in a month.

Formula: Profit = income – expenses (4)

3.1.8 Use any method to calculate the profit that John will make if he sells 110 smartphones in a month.

Formula: Profit = income – expenses (5)

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3.2 John plans to hire temporary sales representatives to assist with 24-hour online sales during the busy December period. The table below shows the number of sales representatives and the number of shifts needed during the month of December.

Number of sales representatives	1	2	3	5	10
Number of shifts	30	15	10	**A**	**B**

3.2.1 Determine the constant value in the relationship between the number of sales representatives and the number of shifts. (2)

3.2.2 Determine the missing values of **A** and **B**. (4)

3.2.3 Use the above table to draw a line graph on the grid found on ANSWER SHEET A (attached). (6)

[35]

QUESTION 4

4.1 John kept a record of the number of smartphones that the sales representatives sold during a shift.

TYPE OF SMARTPHONE	**NUMBER SOLD ONLINE**
Samsung	15
Huawei	39
Hisense	12
Xiaomi	8



4.1.1 How many smartphones were sold altogether? (2)

4.1.2 What is the probability that a sales representative will sell a Hisense smartphone during the next shift? Give your answer in simplest fraction form. (3)

4.1.3 What is the probability that a sales representative will sell a Samsung smartphone during the next shift? Give your answer as a percentage. (3)

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4.2 The table below shows the top five smartphone vendors in China who ship smartphones globally. The table shows the global volume of shipment, market volume and the year-over-year growth of smartphone shipments. The units of the volumes of shipment are in millions.

Top five smartphone vendors in China: Shipments, Market Sales and Year-Over-Year Growth

Vendor	2016 Shipment Volume units in millions	2016 % Market Volume	2015 Shipment Volume units in millions	2015 % Market Volume	Year-Over-Year Growth
OPPO	78,4	16,8%	35,3	8,2%	122,1%
Huawei	76,6	16,4%	62,9	14,6%	21,8%
Vivo	69,2	14,8%	35,1	8,2%	97,2%
Apple	44,9	9,6%	58,4	13,6%	- 23,1%
Xiaomi	41,5	8,9%	64,9	15,1%	- 36,1%
Others	156,7	33,5%	173,3	40,3%	- 9,6%
Total	467,3	100%	429,9	100%	8,7%

[Source: Adapted from IDC Quarterly Mobile Phone Tracker, February 6, 2017]

Study the above table and answer the following questions:

4.2.1 How many units of OPPO smartphones were shipped in 2015? (2)

4.2.2 How many more units of Huawei smartphones were shipped in 2016 than in 2015? (3)

4.2.3 Which smartphone vendor had the lowest year-on-year growth on shipment between 2015 and 2106? (2)

4.2.4 Show how the market volume of 13,6% for the shipment of Apple smartphones in 2015 was calculated. (2)

4.2.5 Show how the year-on-year growth of - 36,1% for the shipment of Xiaomi smartphones was calculated. (4)

4.2.6 Calculate the difference between the medians of the shipment volumes for 2016 and 2015. (6)

4.2.7 Samsung, LG and Sony are top brands in South Africa. Give a reason why you think they do not appear on the table. (1)

4.2.8 Use the data in the year-over-year growth column in the above table to draw a single-bar graph on the grid found on the ANSWER SHEET B (attached). Provide a suitable heading for the graph. (7)

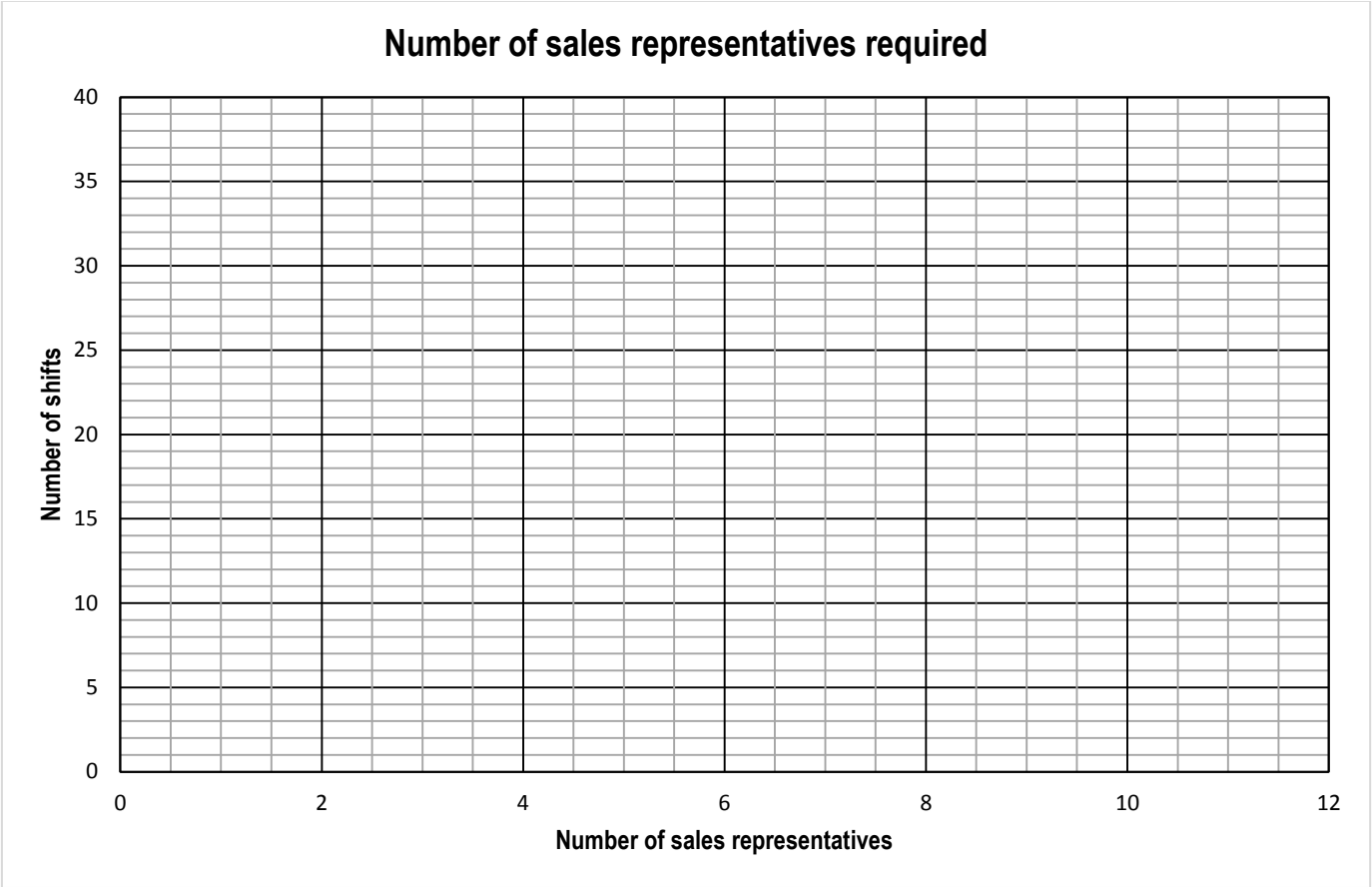
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TOTAL: 150

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ANSWER SHEET A EXAMINATION NUMBER:

QUESTION 3.2.3

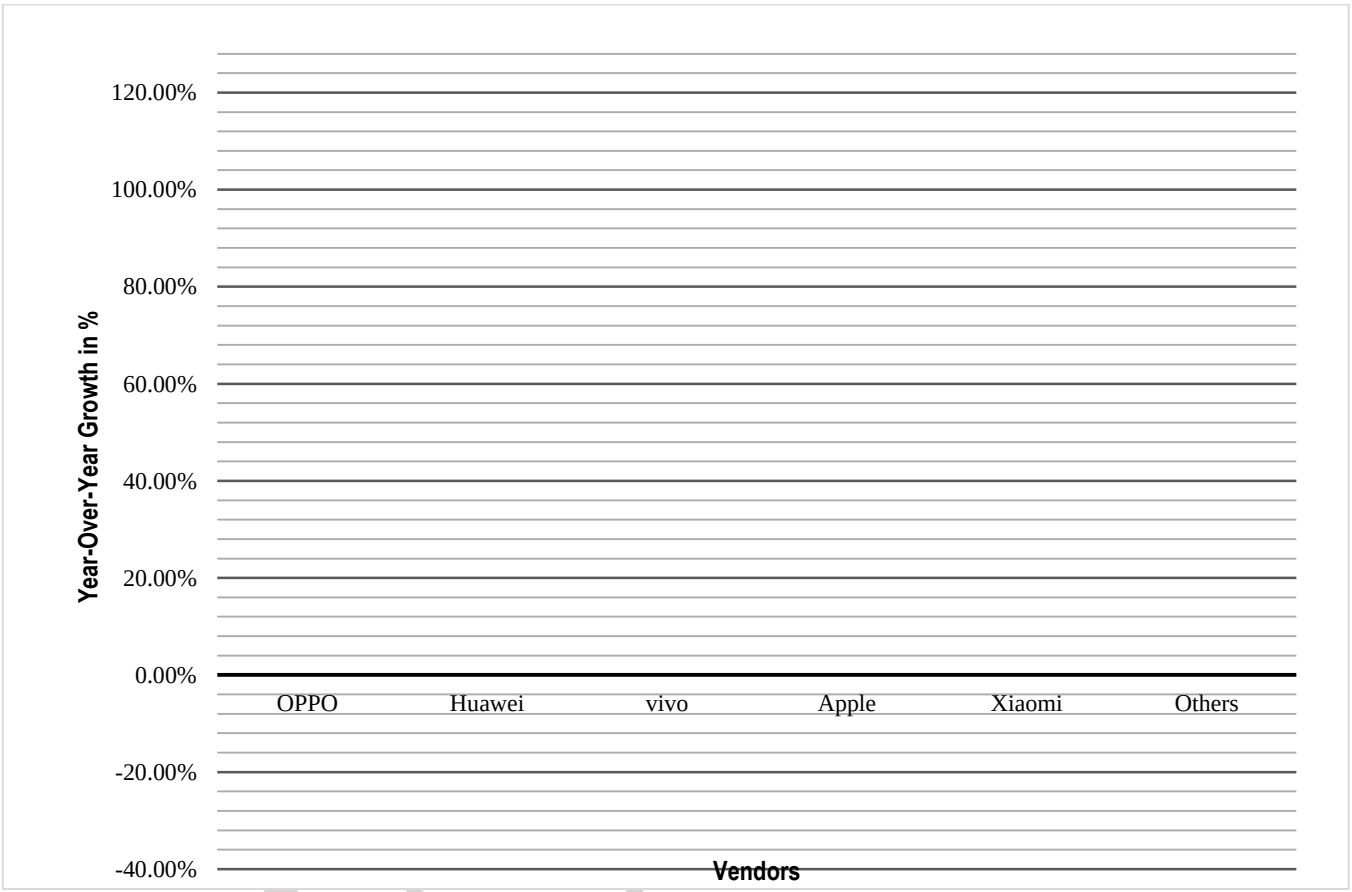


DETACH AND SUBMIT WITH THE ANSWER BOOK.

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ANSWER SHEET B EXAMINATION NUMBER:

QUESTION 4.8



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